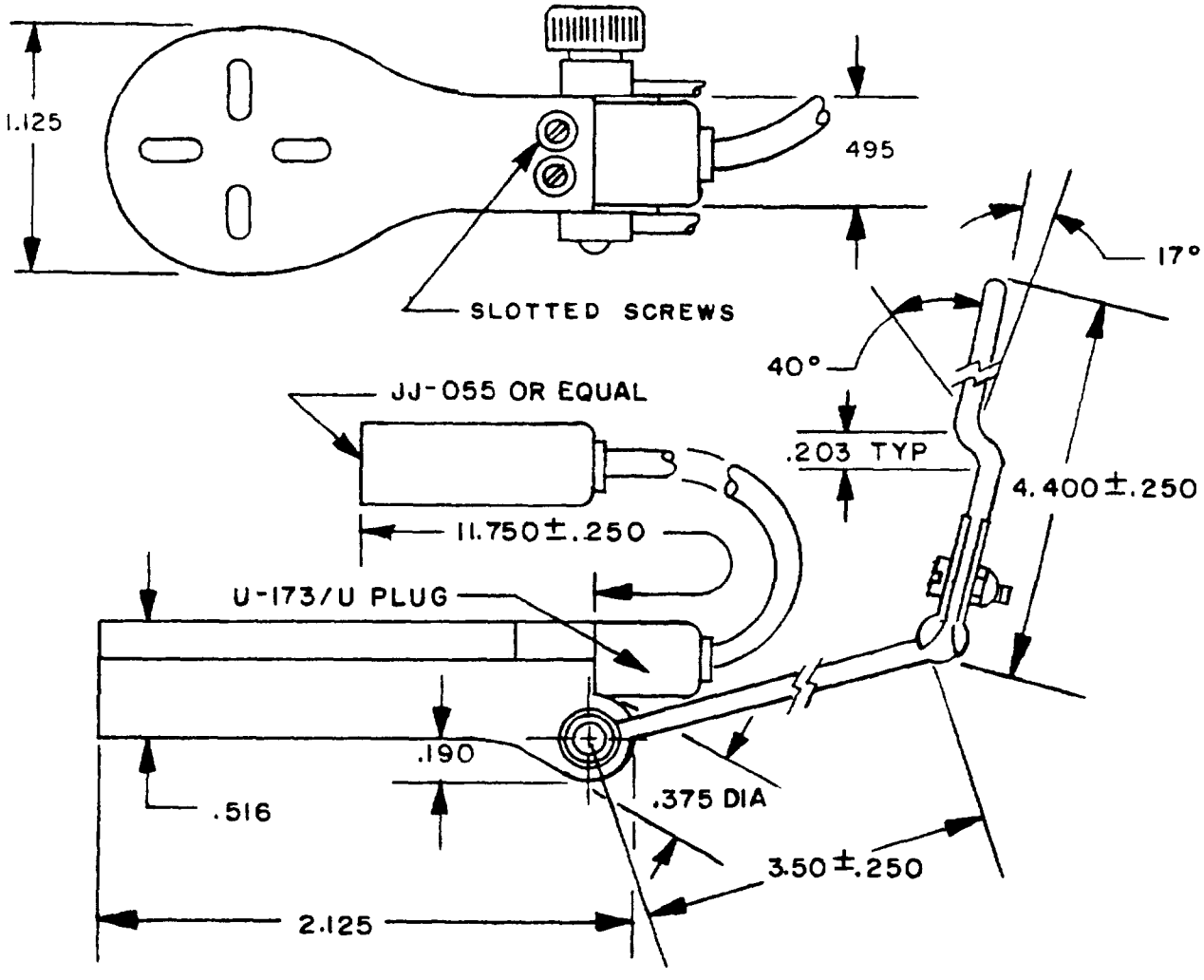
MILITARY SPECIFICATION SHEET

MICROPHONE, M-138/G

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the microphone described herein shall consist of this specification and the latest issue of MIL-M-26542.

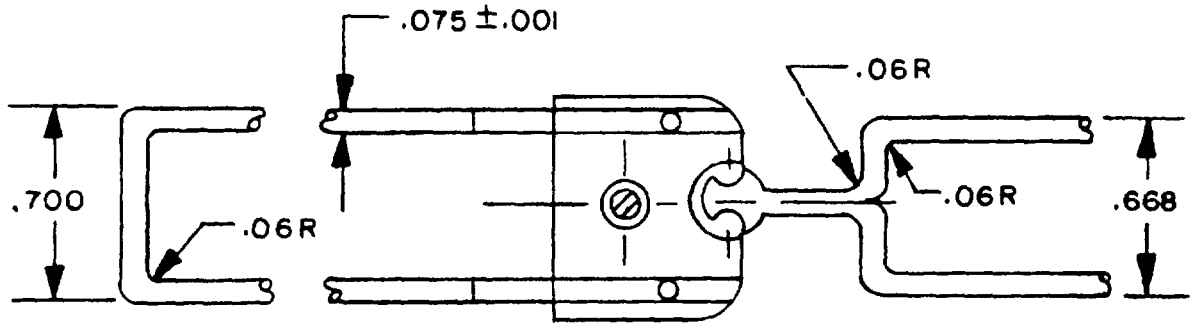


Ⓐ FIGURE 1. Microphone assembly.

Ⓐ denotes changes

FSC 5965

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TOP VIEW OF BALL AND SOCKET JOINT

INCHES	MM	INCHES	MM
.001	0.03	.495	12.57
.06	1.5	.516	13.11
.075	1.91	.668	16.97
.190	4.83	.700	17.78
.203	5.16	1.125	28.58
.250	6.35	2.125	53.96
.350	8.89	4.400	111.76
.375	9.53	11.750	298.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.015 (0.38 mm).

Ⓐ FIGURE 1. Microphone assembly - Continued.

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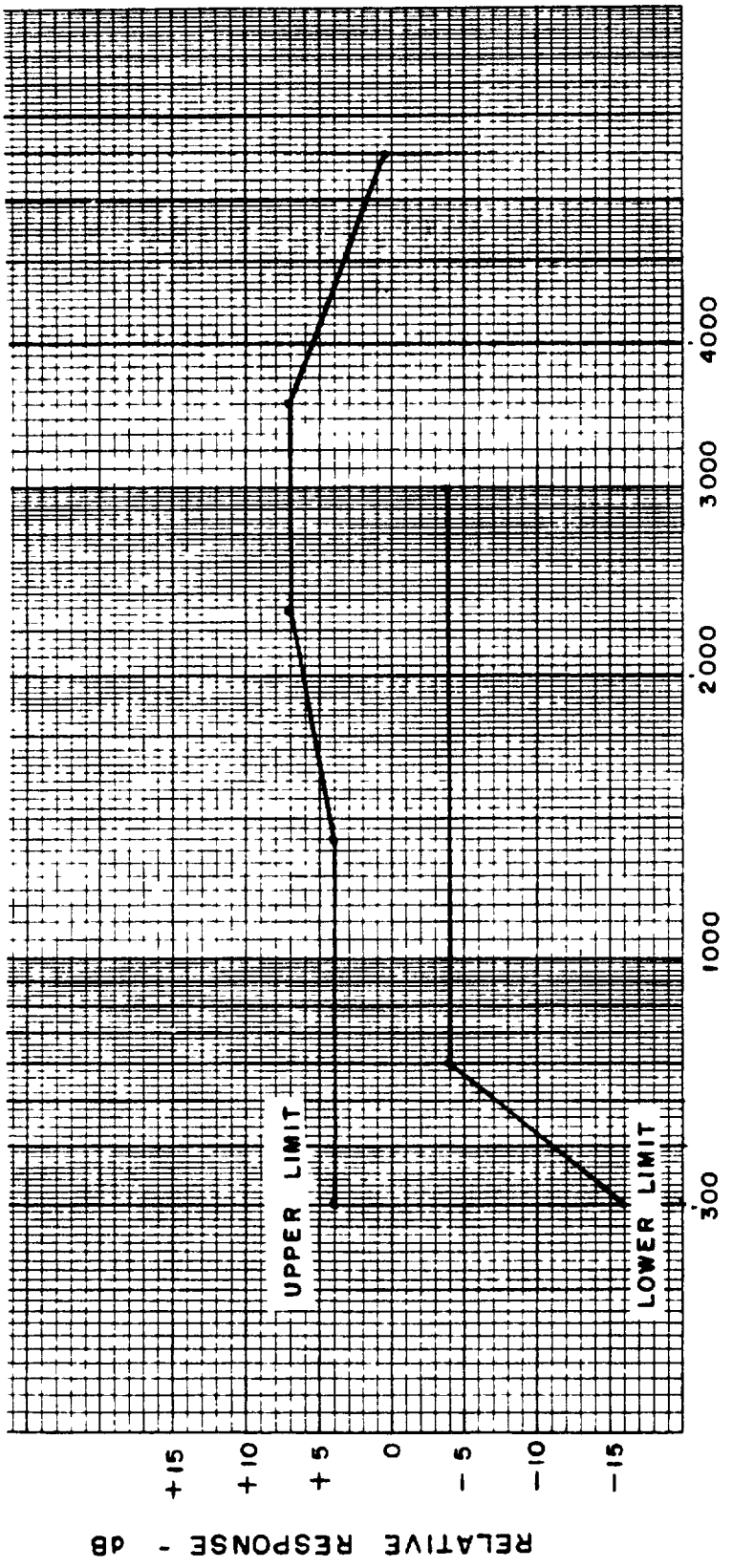


FIGURE 2. Frequency response.

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REQUIREMENTS

Color Semigloss black, 27038 in accordance with FED-STD-595.

Component parts.

Microphone element See figure 1.

Boom. See figure 1.

(A) Cord assembly M22442/38-1.

Plug assembly U-173/U microphone end, JJ-055 type or equal on the free end.

Performance

Sensitivity at ground level: -56 dBm at 1,000 Hz with an input SPL of 28 dynes/cm^2.

(A) Sensitivity at altitude Sensitivity shall be within 3 dB relative to the ground level sensitivity when tested at 15,000 ft.

Frequency response at ground level See figure 2.

Impedance 150 ohms ±15 ohms.

Resistive load 150 ohms.

The microphone assembly shall be tested in accordance with the tests listed in table I.

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Ⓐ TABLE I. Parameter applicability.

Inspection	Requirements paragraph	Test paragraph	Qualification tests	Group "A" tests	Group "B" tests	Group "C" tests
Group I						
Visual and mechanical inspection	3.1,3.4,3.5	4.7.1	X	X		
Sensitivity of ground level	3.5.1	4.7.2.2	X	X		
Sensitivity at altitude	3.5.2	4.7.2.3				
Frequency response at ground level	3.5.3	4.7.2.4	X	X		
Frequency response at altitude	3.5.4	4.7.2.5				
Impedance	3.5.5	4.7.3	X	X		
Noise cancellation characteristics	3.5.7.1	3.7.5.1				
Effect of external magnetic field	3.5.8	4.7.6				
Stray magnetic field	3.5.9	4.7.7				
Linearity	3.5.10	4.7.8				
Talk out	3.5.11	4.7.9				
Dielectric withstanding voltage	3.5.12	4.7.10	X		X	
Signal to noise	3.5.7	4.7.5	X		X	
Distortion	3.5.6	4.7.4	X		X	
Interchangeability	3.5.13	4.7.11	X			
Group II						
Thermal shock	3.5.14	4.7.12	X			X
Humidity	3.5.15	4.7.13	X			X
Drop	3.5.16	4.7.14	X			X
Pressure equalization	3.5.17	4.7.15				
Explosive decompression	3.5.18	4.7.16				
Salt fog	3.5.19	4.7.17	X			X
Group III						
Vibration	3.5.20	4.7.18	X			X
Bounce	3.5.21	4.7.19	X			X
Altitude	3.5.22	4.7.20	X			X
Moisture barrier	3.5.23	4.7.21				
Immersion	3.5.24	4.7.22	X			X
Group IV						
Fungus	3.5.25	4.7.23				
Group V						
Gun blast	3.5.26	4.7.24	X			

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities.
Navy - AS
Air Force - 99
DLA - ES

User activities:
Army - AR, AT, AV, ME
Navy - OS

Agent.
DLA - ES

Preparing activity:
Air Force - 85

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